

Modal Logic: Tableaux Examples

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Example 1:

$$\Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B) \text{ in K}$$

Example 1: $\Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ in K

1. $\models 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ Assumption

Start by assuming it's false.

1. $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ ✓ Assumption
2. $\mathbb{T} 1, \Diamond(A \wedge B) \rightarrow \mathbb{F} 1$
3. $\mathbb{F} 1, \Diamond A \wedge \Diamond B \rightarrow \mathbb{F} 1$

False conditional means true antecedent and false consequent.

- | | | |
|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond(A \wedge B)$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond A \wedge \Diamond B$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, A \wedge B$ | $\Diamond \mathbb{T} 2$ |

True $\Diamond$ introduces world 1.1.

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Diamond(A \wedge B) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond B$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, A \wedge B \checkmark$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\wedge \mathbb{T} 4$
6.	$\mathbb{T} 1.1, B$	$\wedge \mathbb{T} 4$

True conjunction gives us both A and B at 1.1.

1.	$\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B) \checkmark$	Assumption
2.	$\top 1, \Diamond(A \wedge B) \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond A \wedge \Diamond B \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\top 1.1, A \wedge B \checkmark$	$\Diamond \top 2$
5.	$\top 1.1, A$	$\wedge \top 4$
6.	$\top 1.1, B$	$\wedge \top 4$
$\swarrow \quad \searrow$		
7.	$\mathbb{F} 1, \Diamond A \checkmark \quad \mathbb{F} 1, \Diamond B \checkmark$	$\wedge \mathbb{F} 3$
8.	$\mathbb{F} 1.1, A$	$\Diamond \mathbb{F} 7$
9.	$\otimes \quad \mathbb{F} 1.2, B$	$\Diamond \mathbb{F} 7$
	$\otimes$	

When we introduce worlds that make the $\Diamond$ sentences true, both branches close. So this is valid in **K**.